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3 Convex Functions

3.1 Definition and Basic Properties

Definition 3.1 (Convex Function)**.** A function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is *convex* if $\operatorname{dom} f$ is a convex set and for all $x, y \in \operatorname{dom} f$ and $\theta \in [0, 1]$:

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y).$$

The function is *strictly convex* if the inequality is strict whenever $x \neq y$ and $\theta \in (0, 1)$. It is *concave* if $-f$ is convex.

Definition 3.2 (Epigraph)**.** The *epigraph* of $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is

$$\operatorname{epi} f = \{(x, t) \in \mathbb{R}^{n+1} : x \in \operatorname{dom} f,\ f(x) \leq t\}.$$

Proposition 3.3 (Epigraph Characterization)**.** *A function f is convex if and only if* $\operatorname{epi} f$ *is a convex set.*

Proof. ($\Rightarrow$) Let $(x_1, t_1), (x_2, t_2) \in \operatorname{epi} f$ and $\theta \in [0, 1]$. Then $x_1, x_2 \in \operatorname{dom} f$, $f(x_1) \leq t_1$, $f(x_2) \leq t_2$. By convexity of $\operatorname{dom} f$, $\theta x_1 + (1-\theta)x_2 \in \operatorname{dom} f$, and

$$f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta)f(x_2) \leq \theta t_1 + (1-\theta)t_2.$$

Hence $(\theta x_1 + (1-\theta)x_2,\ \theta t_1 + (1-\theta)t_2) \in \operatorname{epi} f$.

($\Leftarrow$) Suppose $\operatorname{epi} f$ is convex. For $x_1, x_2 \in \operatorname{dom} f$ and $\theta \in [0, 1]$: $(x_1, f(x_1))$ and $(x_2, f(x_2))$ lie in $\operatorname{epi} f$, so their convex combination $(\theta x_1 + (1-\theta)x_2,\ \theta f(x_1) + (1-\theta)f(x_2))$ lies in $\operatorname{epi} f$. This means $\theta x_1 + (1-\theta)x_2 \in \operatorname{dom} f$ and $f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta)f(x_2)$. □

Example 3.4 (Standard Convex Functions)**.**

(a) **Affine**: $f(x) = a^T x + b$ is both convex and concave.

(b) **Norms**: Every norm $\|\cdot\|$ on $\mathbb{R}^n$ is convex (by the triangle inequality and homogeneity).

(c) **Quadratic**: $f(x) = x^T P x + q^T x + r$ is convex if and only if $P \succeq 0$.

(d) **Exponential**: $f(x) = e^{ax}$ is convex on $\mathbb{R}$ for any a.

(e) **Negative entropy**: $f(x) = x \log x$ is convex on $\mathbb{R}_{++}$.

(f) **Log-sum-exp**: $f(x) = \log(\sum_{i=1}^n e^{x_i})$ is convex on $\mathbb{R}^n$.

3.2 Restriction to a Line

Proposition 3.5 (Line Restriction)**.** *A function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is convex if and only if for every $x \in \operatorname{dom} f$ and every $v \in \mathbb{R}^n$, the function $g(t) = f(x + tv)$ is convex (on its domain $\{t : x + tv \in \operatorname{dom} f\}$).*

Proof. ($\Rightarrow$) For t_1, t_2 in the domain of g and $\theta \in [0, 1]$:

$$g(\theta t_1 + (1-\theta)t_2) = f(\theta(x + t_1 v) + (1-\theta)(x + t_2 v)) \leq \theta f(x + t_1 v) + (1-\theta)f(x + t_2 v) = \theta g(t_1) + (1-\theta)g(t_2).$$

($\Leftarrow$) For $x_1, x_2 \in \operatorname{dom} f$, set $x = x_1$, $v = x_2 - x_1$, so $g(t) = f(x_1 + t(x_2 - x_1))$, with $g(0) = f(x_1)$, $g(1) = f(x_2)$. Convexity of g gives $g(\theta) \leq \theta g(1) + (1-\theta)g(0) = \theta f(x_2) + (1-\theta)f(x_1)$. □

3.3 First-Order Condition

Theorem 3.6 (First-Order Condition for Convexity). *Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable with convex domain. Then f is convex if and only if for all $x, y \in \operatorname{dom} f$:*

$$f(y) \geq f(x) + \nabla f(x)^T (y - x). \tag{1}$$

Proof. ($\Rightarrow$) Assume f is convex. For $x, y \in \operatorname{dom} f$ and $t \in (0, 1]$:

$$f(x + t(y - x)) \leq (1 - t)f(x) + tf(y).$$

Rearranging: $f(y) \geq f(x) + \frac{f(x+t(y-x))-f(x)}{t}$. Taking $t \to 0^+$, the right-hand side becomes $f(x) + \nabla f(x)^T (y - x)$ by the definition of the directional derivative.

($\Leftarrow$) Assume (1) holds for all x, y. Let $x_1, x_2 \in \operatorname{dom} f$, $\theta \in [0, 1]$, and $z = \theta x_1 + (1 - \theta)x_2$. Apply (1) with $x = z$:

$$f(x_1) \geq f(z) + \nabla f(z)^T (x_1 - z),$$
$$f(x_2) \geq f(z) + \nabla f(z)^T (x_2 - z).$$

Multiply the first by θ and the second by $(1 - \theta)$ and add:

$$\theta f(x_1) + (1 - \theta)f(x_2) \geq f(z) + \nabla f(z)^T \big(\theta(x_1 - z) + (1 - \theta)(x_2 - z)\big) = f(z),$$

since $\theta x_1 + (1 - \theta)x_2 - z = 0$. □

Key Result

The first-order condition (1) says the tangent hyperplane at any point is a *global underestimator* of a convex function. This has a powerful consequence: if $\nabla f(x^*) = 0$, then $f(y) \geq f(x^*)$ for all y, so x^* is a global minimizer.

RL/ML Connection

The first-order condition directly implies that **any local minimum of a convex function is a global minimum**. In RL, this is why convex surrogate losses (e.g., the linear approximation in TRPO) can be minimized greedily without fear of local optima. For non-convex objectives (deep RL), this guarantee fails, motivating trust region and proximal methods to control the update step.

3.4 Second-Order Condition

Theorem 3.7 (Second-Order Condition for Convexity). *Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice differentiable with open convex domain. Then f is convex if and only if the Hessian is positive semidefinite everywhere:*

$$\nabla^2 f(x) \succeq 0 \quad \textit{for all } x \in \operatorname{dom} f.$$

f is strictly convex if $\nabla^2 f(x) \succ 0$ for all x (the converse does not hold: $f(x) = x^4$ is strictly convex but $f''(0) = 0$).

Proof. By the line restriction (Proposition 3.5), it suffices to prove the one-dimensional case: $g : \mathbb{R} \to \mathbb{R}$ twice differentiable is convex if and only if $g''(t) \geq 0$ for all t.

($\Rightarrow$) Suppose g is convex and $g''(t_0) < 0$ for some t_0. By the first-order condition:

$$g(t_0 + h) \geq g(t_0) + g'(t_0)h \quad \text{for all } h.$$

By Taylor's theorem, $g(t_0 + h) = g(t_0) + g'(t_0)h + \frac{1}{2}g''(\xi)h^2$ for some ξ between t_0 and $t_0 + h$. For sufficiently small $|h|$, $g''(\xi) < 0$ by continuity, so $g(t_0 + h) < g(t_0) + g'(t_0)h$, contradicting the first-order condition.

($\Leftarrow$) Suppose $g''(t) \geq 0$ for all t. By Taylor's theorem, for any x, y:

$$g(y) = g(x) + g'(x)(y - x) + \tfrac{1}{2}g''(\xi)(y - x)^2 \geq g(x) + g'(x)(y - x),$$

which is the first-order condition. By Theorem 3.6, g is convex.

For the multivariate case: f is convex iff $g(t) = f(x + tv)$ is convex for all x, v. We have $g''(t) = v^T \nabla^2 f(x + tv)v$. Thus $g''(t) \geq 0$ for all t, v iff $\nabla^2 f \succeq 0$ everywhere. □

Example 3.8 (Quadratic Functions)**.** For $f(x) = \frac{1}{2}x^T Px + q^T x + r$, we have $\nabla^2 f(x) = P$. So f is convex iff $P \succeq 0$, and strictly convex iff $P \succ 0$.

Example 3.9 (Log-Sum-Exp)**.** For $f(x) = \log \sum_{i=1}^n e^{x_i}$, the Hessian is

$$\nabla^2 f(x) = \frac{1}{\mathbf{1}^T z} \operatorname{diag}(z) - \frac{1}{(\mathbf{1}^T z)^2} zz^T,$$

where $z_i = e^{x_i}$. For any v:

$$v^T \nabla^2 f(x) v = \frac{(\sum z_i)(\sum z_i v_i^2) - (\sum z_i v_i)^2}{(\sum z_i)^2} \geq 0,$$

by the Cauchy–Schwarz inequality (with weights z_i). Hence f is convex.

3.5 Jensen's Inequality

Theorem 3.10 (Jensen's Inequality)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and let X be a random vector with $\mathbb{E}[\|X\|] < \infty$ and $\mathbb{E}[X] \in \operatorname{dom} f$. Then*

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

If f is concave, the inequality reverses.

Proof. By the first-order condition (Theorem 3.6) applied at $x = \mathbb{E}[X]$:

$$f(X) \geq f(\mathbb{E}[X]) + \nabla f(\mathbb{E}[X])^T (X - \mathbb{E}[X]).$$

Taking expectations:

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]) + \nabla f(\mathbb{E}[X])^T (\mathbb{E}[X] - \mathbb{E}[X]) = f(\mathbb{E}[X]).$$

For the non-differentiable case, use the existence of a subgradient $g \in \partial f(\mathbb{E}[X])$ satisfying $f(y) \geq f(\mathbb{E}[X]) + g^T(y - \mathbb{E}[X])$ for all y (subgradients exist for all convex functions at interior points of their domain). The same argument applies. □

Jensen's Inequality – Finite Form

For convex f, points $x_1, \ldots, x_k$, and weights $\theta_i \geq 0$ with $\sum \theta_i = 1$:

$$f\left(\sum_{i=1}^k \theta_i x_i\right) \leq \sum_{i=1}^k \theta_i f(x_i).$$

This is the defining property of convexity extended from two points to k points by induction.

RL/ML Connection

Jensen's inequality is ubiquitous in RL/ML analysis:

- **Evidence lower bound (ELBO)**: Since log is concave, $\log \mathbb{E}[f(X)] \geq \mathbb{E}[\log f(X)]$, which is the foundation of variational inference.
- **KL divergence**: Non-negativity of $D_{\mathrm{KL}}(p\|q) = \mathbb{E}_p[\log(p/q)] \geq 0$ follows from Jensen applied to $-\log$ (convex).
- **Importance sampling**: Variance analysis of importance-weighted estimators uses Jensen to bound $\mathbb{E}[(w(X)f(X))^2]$ terms.

4 Operations Preserving Convexity of Functions

4.1 Nonnegative Weighted Sums

Proposition 4.1 (Nonnegative Weighted Sum). *If $f_1, \ldots, f_m$ are convex and $w_1, \ldots, w_m \geq 0$, then $f = \sum_{i=1}^m w_i f_i$ is convex.*

Proof. For $\theta \in [0,1]$:

$$f(\theta x+(1-\theta)y) = \sum_i w_i f_i(\theta x+(1-\theta)y) \leq \sum_i w_i[\theta f_i(x)+(1-\theta)f_i(y)] = \theta f(x)+(1-\theta)f(y). \quad \square$$

4.2 Composition with Affine Maps

Proposition 4.2 (Affine Composition). *If $f : \mathbb{R}^m \to \mathbb{R}$ is convex, $A \in \mathbb{R}^{m\times n}$, and $b \in \mathbb{R}^m$, then $g(x) = f(Ax + b)$ is convex.*

Proof. $\operatorname{dom} g = \{x : Ax + b \in \operatorname{dom} f\}$ is convex (preimage of convex set under affine map). For $\theta \in [0,1]$:

$$\begin{aligned} g(\theta x + (1-\theta)y) &= f(A(\theta x + (1-\theta)y) + b) \\ &= f(\theta(Ax + b) + (1-\theta)(Ay + b)) \\ &\leq \theta f(Ax + b) + (1-\theta)f(Ay + b) = \theta g(x) + (1-\theta)g(y). \end{aligned} \quad \square$$

4.3 Pointwise Maximum and Supremum

Proposition 4.3 (Pointwise Maximum). *If $f_1, \ldots, f_m$ are convex, then $f(x) = \max\{f_1(x), \ldots, f_m(x)\}$ is convex.*

Proof. For $\theta \in [0,1]$:

$$f(\theta x+(1-\theta)y) = \max_i f_i(\theta x+(1-\theta)y) \leq \max_i[\theta f_i(x)+(1-\theta)f_i(y)] \leq \theta \max_i f_i(x)+(1-\theta)\max_i f_i(y).$$

The last inequality uses: $\max_i(a_i + b_i) \leq \max_i a_i + \max_i b_i$ (applied with $a_i = \theta f_i(x)$ and $b_i = (1-\theta)f_i(y)$, noting $\theta \geq 0$ and $1-\theta \geq 0$). $\square$

Proposition 4.4 (Pointwise Supremum). *If $f(x, y)$ is convex in x for each $y \in \mathcal{A}$, then $g(x) = \sup_{y\in\mathcal{A}} f(x, y)$ is convex (with $\operatorname{dom} g = \{x : g(x) < +\infty\}$).*

Proof. The epigraph of g is $\operatorname{epi} g = \bigcap_{y\in\mathcal{A}} \operatorname{epi} f(\cdot, y)$. Each $\operatorname{epi} f(\cdot, y)$ is convex (since $f(\cdot, y)$ is convex), and the intersection of convex sets is convex (Proposition **??**). Hence $\operatorname{epi} g$ is convex, so g is convex (Proposition 3.3). $\square$

RL/ML Connection

The pointwise supremum rule shows why the **support function** $S_C(y) = \sup_{x \in C} y^T x$ is convex for any set C (not necessarily convex). In RL, the optimal value function $V^*(s) = \sup_\pi V^\pi(s)$ is the supremum of affine functions of the transition model parameters in certain formulations, connecting to the convexity of the value function in the occupancy measure.

4.4 Composition Rules

Proposition 4.5 (Scalar Composition). *Let $h : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}$. Define $f = h \circ g$. Then:*

(a) f is convex if h is convex and nondecreasing, and g is convex.

(b) f is convex if h is convex and nonincreasing, and g is concave.

(c) f is concave if h is concave and nondecreasing, and g is concave.

(d) f is concave if h is concave and nonincreasing, and g is convex.

Proof. We prove (a); the others are analogous. Assume h, g twice differentiable for clarity. Then $f''(x) = h''(g(x))(g'(x))^2 + h'(g(x))g''(x)$ (in 1D). Since h is convex, $h'' \geq 0$; since h is nondecreasing, $h' \geq 0$; since g is convex, $g'' \geq 0$. Both terms are nonnegative, so $f'' \geq 0$.

For the general (non-differentiable) proof of (a): let $\theta \in [0, 1]$. Since g is convex: $g(\theta x + (1 - \theta)y) \leq \theta g(x) + (1 - \theta)g(y)$. Since h is nondecreasing:

$$h(g(\theta x + (1 - \theta)y)) \leq h(\theta g(x) + (1 - \theta)g(y)).$$

Since h is convex:

$$h(\theta g(x) + (1 - \theta)g(y)) \leq \theta h(g(x)) + (1 - \theta)h(g(y)).$$

Combining: $f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$. □

Example 4.6 (Applications of Composition Rules). (a) $e^{g(x)}$ is convex if g is convex ($h(t) = e^t$ is convex and nondecreasing).

(b) $1/g(x)$ is convex if g is concave and positive ($h(t) = 1/t$ is convex and nonincreasing on $\mathbb{R}_{++}$).

(c) $\log g(x)$ is concave if g is concave and positive ($h(t) = \log t$ is concave and nondecreasing).

(d) $\|Ax + b\|_p$ is convex ($h = \|\cdot\|_p$ is convex and nondecreasing on the nonnegative orthant is not needed here; use affine composition Proposition 4.2 directly since norms are convex).

4.5 Infimum / Minimization

Proposition 4.7 (Partial Minimization). *If $f(x, y)$ is convex in (x, y) and C is a convex set, then $g(x) = \inf_{y \in C} f(x, y)$ is convex (provided $g(x) > -\infty$ for all x).*

Proof. Let $x_1, x_2 \in \operatorname{dom} g$ and $\theta \in [0, 1]$. For any $\varepsilon > 0$, choose $y_1, y_2 \in C$ such that $f(x_i, y_i) \leq g(x_i) + \varepsilon$. Then

$$\begin{aligned} g(\theta x_1 + (1 - \theta)x_2) &\leq f(\theta x_1 + (1 - \theta)x_2,\ \theta y_1 + (1 - \theta)y_2) \\ &\leq \theta f(x_1, y_1) + (1 - \theta)f(x_2, y_2) \\ &\leq \theta g(x_1) + (1 - \theta)g(x_2) + \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, $g(\theta x_1 + (1 - \theta)x_2) \leq \theta g(x_1) + (1 - \theta)g(x_2)$. □

4.6 Perspective Transform

Proposition 4.8 (Perspective of a Convex Function)**.** *If $f : \mathbb{R}^n \to \mathbb{R}$ is convex, then its* perspective *$g : \mathbb{R}^{n+1} \to \mathbb{R}$ defined by $g(x,t) = tf(x/t)$ with $\operatorname{dom} g = \{(x,t) : x/t \in \operatorname{dom} f,\ t > 0\}$ is convex.*

Proof. We show $\operatorname{epi} g$ is convex. Note $(x,t,s) \in \operatorname{epi} g$ iff $t > 0$ and $tf(x/t) \le s$, iff $f(x/t) \le s/t$, iff $(x/t, s/t) \in \operatorname{epi} f$. Thus $(x,t,s) \in \operatorname{epi} g$ iff $(x/t, s/t) \in \operatorname{epi} f$ and $t > 0$. This is the preimage of $\operatorname{epi} f$ under the perspective map $(x,t,s) \mapsto (x/t, s/t)$. By Proposition **??**, the perspective map preserves convexity of preimages. Since $\operatorname{epi} f$ is convex (as f is convex), $\operatorname{epi} g$ is convex. □

Example 4.9 (KL Divergence from Perspective)**.** The negative entropy $f(x) = x \log x$ is convex on $\mathbb{R}_{++}$. Its perspective is $g(x,t) = t \cdot (x/t)\log(x/t) = x\log(x/t)$. Summing over coordinates gives the KL divergence: $D_{\mathrm{KL}}(p\|q) = \sum_i p_i \log(p_i/q_i)$, which is jointly convex in (p,q).

5 Conjugate Functions

5.1 The Fenchel Conjugate

Definition 5.1 (Fenchel Conjugate)**.** Let $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$. The *Fenchel conjugate* (or *convex conjugate*) of f is $f^* : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ defined by

$$f^*(y) = \sup_{x \in \operatorname{dom} f} \left(y^T x - f(x)\right).$$

Proposition 5.2 (Conjugate is Always Convex)**.** *For any function f (not necessarily convex), f^* is convex and closed (i.e., lower semicontinuous).*

Proof. For each fixed x, the function $y \mapsto y^T x - f(x)$ is affine in y, hence convex. The conjugate $f^*(y) = \sup_x (y^T x - f(x))$ is the pointwise supremum of convex (in fact affine) functions of y. By Proposition 4.4, f^* is convex. Closedness follows because $\operatorname{epi} f^* = \bigcap_x \{(y,t) : y^T x - f(x) \le t\}$ is an intersection of closed halfspaces. □

Theorem 5.3 (Fenchel's Inequality)**.** *For any function f and any $x, y \in \mathbb{R}^n$:*

$$f(x) + f^*(y) \ge y^T x.$$

Proof. By definition, $f^*(y) = \sup_z (y^T z - f(z)) \ge y^T x - f(x)$. Rearranging gives $f(x) + f^*(y) \ge y^T x$. □

Fenchel's Inequality

The inequality $f(x) + f^*(y) \ge \langle y, x\rangle$ is often called the *Young–Fenchel inequality*. Equality holds if and only if $y \in \partial f(x)$ (the subdifferential of f at x), when f is convex and closed.

5.2 Conjugate of the Conjugate

Theorem 5.4 (Biconjugate Theorem)**.** *If $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is convex and closed (lower semicontinuous) with $f \not\equiv +\infty$, then $f^{**} = f$.*

Proof sketch. By Fenchel's inequality, $f^{**}(x) = \sup_y (x^T y - f^*(y)) \le f(x)$ for all x (set $z = x$ in the supremum defining f^*, then rearrange). The reverse inequality $f^{**} \ge f$ uses the supporting hyperplane theorem: since $\operatorname{epi} f$ is a closed convex set and $(x, f(x))$ lies on its boundary, there is a supporting hyperplane $a^T u + bt \ge a^T x + bf(x)$ for all $(u,t) \in \operatorname{epi} f$. Analyzing this gives $f^*(-a/b) = (-a/b)^T x - f(x)$, hence $f^{**}(x) \ge (-a/b)^T x - f^*(-a/b) = f(x)$. A complete proof requires careful handling of the case $b = 0$ and uses the full Fenchel–Moreau theorem from convex analysis. □

5.3 Examples of Conjugates

Example 5.5 (Conjugate of a Linear Function)**.** Let $f(x) = a^T x + b$. Then

$$f^*(y) = \sup_x (y^T x - a^T x - b) = \sup_x ((y-a)^T x) - b = \begin{cases} -b & \text{if } y = a, \\ +\infty & \text{otherwise.} \end{cases}$$

Example 5.6 (Conjugate of a Quadratic)**.** Let $f(x) = \frac{1}{2} x^T Q x$ where $Q \succ 0$. Then

$$f^*(y) = \sup_x \left(y^T x - \tfrac{1}{2} x^T Q x \right).$$

Setting the gradient to zero: $y - Qx = 0$, so $x^* = Q^{-1} y$. Substituting:

$$f^*(y) = y^T Q^{-1} y - \tfrac{1}{2} y^T Q^{-1} Q Q^{-1} y = \tfrac{1}{2} y^T Q^{-1} y.$$

Example 5.7 (Conjugate of the Negative Entropy)**.** Let $f(x) = x \log x$ for $x > 0$. Then $f^*(y) = \sup_{x>0}(yx - x \log x)$. Setting the derivative to zero: $y - \log x - 1 = 0$, so $x^* = e^{y-1}$. Thus $f^*(y) = e^{y-1}$.

Example 5.8 (Conjugate of Indicator Function)**.** Let $\delta_C(x) = 0$ if $x \in C$, $+\infty$ otherwise. Then

$$\delta_C^*(y) = \sup_{x \in C} y^T x = S_C(y),$$

the *support function* of C. This connects set geometry to duality.

Example 5.9 (Conjugate of Norm)**.** Let $f(x) = \|x\|$. Then $f^*(y) = \delta_{\{y : \|y\|_* \le 1\}}(y)$, where $\|y\|_* = \sup_{\|x\| \le 1} y^T x$ is the dual norm. To see this: $f^*(y) = \sup_x (y^T x - \|x\|)$. If $\|y\|_* > 1$, then there exists x_0 with $\|x_0\| = 1$ and $y^T x_0 > 1$, so $f^*(y) \ge \sup_{t \ge 0}(t y^T x_0 - t) = +\infty$. If $\|y\|_* \le 1$, then $y^T x \le \|y\|_* \|x\| \le \|x\|$, so $f^*(y) \le 0$; and $f^*(y) \ge y^T \cdot 0 - \|0\| = 0$. Thus $f^*(y) = 0$.

RL/ML Connection

Fenchel conjugates appear naturally in **regularized MDPs**. If the policy is regularized by an entropy term $\Omega(\pi) = \sum_a \pi(a) \log \pi(a)$, the regularized Bellman operator involves Ω^*, the conjugate of the entropy, which equals the log-sum-exp function (the "soft maximum"). This is the foundation of **soft Q-learning** and **SAC (Soft Actor-Critic)**, where the softmax policy arises as the conjugate dual solution.

6 Sublevel Sets and Quasiconvex Functions

6.1 Sublevel Sets

Definition 6.1 (Sublevel Set)**.** The *α-sublevel set* of $f : \mathbb{R}^n \to \mathbb{R}$ is $C_\alpha = \{x \in \operatorname{dom} f : f(x) \le \alpha\}$.

Proposition 6.2 (Sublevel Sets of Convex Functions)**.** *If f is convex, then all its sublevel sets are convex.*

Proof. Let $x, y \in C_\alpha$ (so $f(x) \le \alpha$ and $f(y) \le \alpha$) and $\theta \in [0, 1]$. Then

$$f(\theta x + (1-\theta) y) \le \theta f(x) + (1-\theta) f(y) \le \theta \alpha + (1-\theta)\alpha = \alpha,$$

so $\theta x + (1-\theta) y \in C_\alpha$. □

Warning

The converse of Proposition 6.2 is **false**. A function can have all sublevel sets convex without being convex. For example, $f(x) = -e^x$ has convex sublevel sets $\{x : -e^x \leq \alpha\} = \{x : x \geq \log(-\alpha)\}$ (halflines) for $\alpha < 0$, but f is concave (not convex). Functions with convex sublevel sets are called *quasiconvex* (see below).

6.2 Quasiconvex Functions

Definition 6.3 (Quasiconvex Function)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is *quasiconvex* if $\operatorname{dom} f$ is convex and all sublevel sets $\{x : f(x) \leq \alpha\}$ are convex. Equivalently, for all $x, y \in \operatorname{dom} f$ and $\theta \in [0, 1]$:

$$f(\theta x + (1-\theta)y) \leq \max\{f(x), f(y)\}.$$

f is *quasiconcave* if $-f$ is quasiconvex, i.e., all *superlevel sets* $\{x : f(x) \geq \alpha\}$ are convex.

Proposition 6.4 (Equivalence of Quasiconvexity Definitions)**.** *f has all sublevel sets convex if and only if $f(\theta x + (1-\theta)y) \leq \max\{f(x), f(y)\}$ for all $x, y \in \operatorname{dom} f$ and $\theta \in [0, 1]$.*

Proof. ($\Rightarrow$) Let $\alpha = \max\{f(x), f(y)\}$. Then $x, y \in C_\alpha$, which is convex, so $\theta x + (1-\theta)y \in C_\alpha$, giving $f(\theta x + (1-\theta)y) \leq \alpha = \max\{f(x), f(y)\}$.

($\Leftarrow$) Fix α and let $x, y \in C_\alpha$. Then $\max\{f(x), f(y)\} \leq \alpha$, so $f(\theta x + (1-\theta)y) \leq \alpha$, giving $\theta x + (1-\theta)y \in C_\alpha$. □

Example 6.5 (Quasiconvex Functions)**.** (a) Every convex function is quasiconvex.

(b) $f(x) = \lceil x \rceil$ (ceiling function) is quasiconvex.

(c) $\log x$ is both quasiconvex and quasiconcave on $\mathbb{R}_{++}$ (it is monotone, and monotone functions are both quasiconvex and quasiconcave).

(d) The ratio $f(x) = p(x)/q(x)$ where p is convex nonneg and q is concave positive is quasiconvex.

Proposition 6.6 (First-Order Condition for Quasiconvexity)**.** *A differentiable function f with convex domain is quasiconvex if and only if for all $x, y \in \operatorname{dom} f$:*

$$f(y) \leq f(x) \implies \nabla f(x)^T (y - x) \leq 0.$$

Proof. ($\Rightarrow$) Suppose f is quasiconvex and $f(y) \leq f(x)$. For $t \in (0, 1]$: $f(x + t(y-x)) \leq \max\{f(x), f(y)\} = f(x)$. So $\frac{f(x+t(y-x))-f(x)}{t} \leq 0$. Taking $t \to 0^+$: $\nabla f(x)^T (y - x) \leq 0$.

($\Leftarrow$) We prove the contrapositive. Suppose f is not quasiconvex, so there exist x, y and $\theta \in (0, 1)$ with $f(z) > \max\{f(x), f(y)\}$ where $z = \theta x + (1-\theta)y$. Then $f(x) < f(z)$ and $f(y) < f(z)$. The condition would require $\nabla f(z)^T (x - z) \leq 0$ and $\nabla f(z)^T (y - z) \leq 0$. But $\theta(x - z) + (1-\theta)(y - z) = 0$, and both inner products nonpositive with a nontrivial linear combination equal to zero forces both to be zero. A more careful argument using the mean value theorem on the line segment shows a contradiction with the strict inequality $f(z) > f(x)$. □

7 Log-Concave and Log-Convex Functions

Definition 7.1 (Log-Concave and Log-Convex Functions)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ with $f(x) > 0$ for all $x \in \operatorname{dom} f$ is *log-concave* if $\log f$ is concave, i.e., for all $x, y \in \operatorname{dom} f$ and $\theta \in [0, 1]$:

$$f(\theta x + (1-\theta)y) \geq f(x)^\theta f(y)^{1-\theta}.$$

f is *log-convex* if $\log f$ is convex, equivalently:

$$f(\theta x + (1-\theta)y) \leq f(x)^\theta f(y)^{1-\theta}.$$

Remark 7.2. The defining inequality $f(\theta x+(1-\theta)y) \geq f(x)^\theta f(y)^{1-\theta}$ follows from exponentiating the concavity of $\log f$:

$$\log f(\theta x + (1-\theta)y) \geq \theta \log f(x) + (1-\theta)\log f(y) = \log\big(f(x)^\theta f(y)^{1-\theta}\big).$$

Proposition 7.3 (Hierarchy of Convexity Notions)**.** *For a positive function f:*

$$\textit{log-convex} \implies \textit{convex} \implies \textit{quasiconvex},$$
$$\textit{concave} \implies \textit{log-concave} \implies \textit{quasiconcave}.$$

Proof. **Log-convex $\implies$ convex**: If $\log f$ is convex, then $f = e^{\log f}$ is the composition of the convex nondecreasing function e^t with the convex function $\log f$, hence convex by Proposition 4.5(a).

Convex $\implies$ quasiconvex: Already proved (sublevel sets of convex functions are convex).

Concave $\implies$ log-concave: We must show $\log f$ is concave, assuming f is concave and positive. By the AM–GM inequality, for $\theta \in [0,1]$:

$$f(\theta x + (1-\theta)y) \geq \theta f(x) + (1-\theta)f(y) \geq f(x)^\theta f(y)^{1-\theta},$$

where the first inequality is concavity of f and the second is the weighted AM–GM inequality. Taking logarithms gives concavity of $\log f$.

Log-concave $\implies$ quasiconcave: If $\log f$ is concave, then $\log f$ has convex superlevel sets (since concave functions have convex superlevel sets). The α-superlevel set of f is $\{x : f(x) \geq \alpha\} = \{x : \log f(x) \geq \log \alpha\}$ (for $\alpha > 0$), which is a superlevel set of the concave function $\log f$, hence convex. □

Example 7.4 (Log-Concave Distributions)**.** Many common probability densities are log-concave:

(a) **Gaussian**: $f(x) \propto \exp(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu))$. Here $\log f$ is a concave quadratic.

(b) **Exponential**: $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. $\log f = \log\lambda - \lambda x$ is concave (affine).

(c) **Uniform**: $f(x) = 1/(b-a)$ on $[a,b]$. $\log f$ is constant (concave).

(d) **Laplace**: $f(x) \propto e^{-|x-\mu|/b}$. $\log f$ is concave (piecewise linear with negative slopes).

Proposition 7.5 (Properties of Log-Concave Functions)**.** *(a) Products: if f, g are log-concave, then fg is log-concave.*

(b) Marginals: if $f(x,y)$ is log-concave, then $g(x) = \int f(x,y)\,dy$ is log-concave (Prekopa's theorem).

(c) Convolution: if f, g are log-concave, then $(f * g)(x) = \int f(x-y)g(y)\,dy$ is log-concave.

Proof. (a) $\log(fg) = \log f + \log g$ is the sum of two concave functions, hence concave.

(b) This is Prekopa's theorem. The proof uses the Prekopa–Leindler inequality: if $f, g, h : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ satisfy $h(\theta x + (1-\theta)y) \geq f(x)^\theta g(y)^{1-\theta}$ for all x, y and some $\theta \in (0,1)$, then

$$\int h \geq \left(\int f\right)^\theta \left(\int g\right)^{1-\theta}.$$

Apply this to the "slices" of the log-concave function in the variable being integrated out. Specifically, for fixed x_1, x_2 and $\bar{x} = \theta x_1 + (1-\theta)x_2$, the log-concavity of f gives $f(\bar{x}, y) \geq f(x_1, y_1)^\theta f(x_2, y_2)^{1-\theta}$ when $y = \theta y_1 + (1-\theta)y_2$. Applying the Prekopa–Leindler inequality in y: $g(\bar{x}) \geq g(x_1)^\theta g(x_2)^{1-\theta}$.

(c) Write $(f * g)(x) = \int f(x-y)g(y)\,dy = \int h(x,y)\,dy$ where $h(x,y) = f(x-y)g(y)$. The function h is log-concave in (x,y) since $\log h(x,y) = \log f(x-y) + \log g(y)$ is the sum of a concave function of (x,y) (composition of concave $\log f$ with affine $(x,y) \mapsto x-y$) and a concave function of y (hence of (x,y)). By part (b), the integral over y is log-concave in x. □

RL/ML Connection

Log-concavity is central to Bayesian inference in machine learning:

- A Gaussian prior times a log-concave likelihood gives a log-concave posterior (by the product rule), guaranteeing unimodality and enabling efficient sampling (e.g., Langevin dynamics converges in polynomial time for log-concave targets).
- The marginal property ensures that marginalizing latent variables from a log-concave joint preserves log-concavity, important for evidence computation.
- In RL, the softmax policy $\pi(a|s) \propto \exp(Q(s,a)/\tau)$ is log-concave (in fact log-linear) in Q, which simplifies the analysis of policy gradient methods.

8 Exercises

Exercise 8.1 (Convexity of Set Operations)**.** Let $C_1, C_2 \subseteq \mathbb{R}^n$ be convex sets. Prove that the following are convex:

(a) The Minkowski sum $C_1 + C_2 = \{x_1 + x_2 : x_1 \in C_1, x_2 \in C_2\}$.

(b) The set $\{(\lambda, x) : x \in \lambda C_1,\ \lambda \geq 0\} \subseteq \mathbb{R}^{n+1}$.

Exercise 8.2 (Convexity via Epigraph)**.** Use the epigraph characterization to prove that $f(x) = \max\{|x_1|, \ldots, |x_n|\}$ (the ℓ_∞-norm) is convex. *Hint*: Express epi f as an intersection of halfspaces.

Exercise 8.3 (First-Order Condition Application)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and differentiable.

(a) Show that $\nabla f(x) = 0$ implies x is a global minimizer.

(b) Show that if f is strictly convex, there is at most one global minimizer.

(c) Give an example of a convex function with no minimizer, a unique minimizer, and a non-unique set of minimizers.

Exercise 8.4 (Conjugate Computation)**.** Compute the Fenchel conjugate of each function:

(a) $f(x) = \|x\|_1$ (the ℓ_1-norm on $\mathbb{R}^n$).

(b) $f(x) = -\log x$ for $x > 0$.

(c) $f(x) = \frac{1}{p}\|x\|_p^p$ for $p > 1$.

Exercise 8.5 (Jensen's Inequality Applications)**.** Using Jensen's inequality:

(a) Prove $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$ for any random variable X (i.e., $\mathrm{Var}(X) \geq 0$).

(b) Prove that for positive weights w_i summing to 1 and positive reals a_i: $\prod_{i=1}^n a_i^{w_i} \leq \sum_{i=1}^n w_i a_i$ (weighted AM–GM).

(c) Prove that $D_{\mathrm{KL}}(p\|q) \geq 0$ for any two probability distributions p, q on a finite set.

Exercise 8.6 (Log-Sum-Exp Properties)**.** Let $\mathrm{LSE}(x) = \log(\sum_{i=1}^n e^{x_i})$.

(a) Verify that $\max_i x_i \leq \mathrm{LSE}(x) \leq \max_i x_i + \log n$.

(b) Show that LSE is the conjugate of the negative entropy restricted to the probability simplex: $\mathrm{LSE}(y) = \sup_{x \in \Delta_n}(y^T x + \sum_i x_i \log x_i)$ where Δ_n is the simplex. *Hint*: Use Lagrange multipliers.

Exercise 8.7 (Perspective Transform and KL Divergence). (a) Verify that the perspective of $f(u) = u^2$ is $g(u,t) = u^2/t$ (for $t > 0$), and confirm g is convex.

(b) Show that the perspective of $f(u) = -\log u$ (for $u > 0$) is $g(u,t) = t\log(t/u)$, and relate this to the KL divergence.

Exercise 8.8 (Quasiconvex but Not Convex). Show that $f(x) = \sqrt{|x|}$ is quasiconvex on $\mathbb{R}$ but not convex. *Hint*: For quasiconvexity, show sublevel sets are intervals. For non-convexity, find a counterexample to the convexity inequality.

Exercise 8.9 (Separating Hyperplane Application). Let $C = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$ (the unit disk) and $p = (2,0)$.

(a) Find a separating hyperplane between C and $\{p\}$.

(b) Find a supporting hyperplane to C at the point $(1,0)$.

(c) Is the supporting hyperplane unique at this point? Prove your answer.

Exercise 8.10 (Convexity Preservation Under Partial Minimization). Let $f(x,y) = x^2+y^2+xy$ defined on $\mathbb{R}^2$.

(a) Show that f is convex by computing the Hessian.

(b) Compute $g(x) = \inf_y f(x,y)$ explicitly.

(c) Verify directly that g is convex.

9 Summary

End of Lesson 8: Convex Analysis

Table 1: Summary of key concepts in convex analysis

Concept	**Definition / Condition**	**Key Property**
Convex set	$\theta x + (1-\theta)y \in C$ for $\theta \in [0,1]$	Closed under intersection, affine maps
Convex function	$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y)$	Epigraph is convex set
First-order cond.	$f(y) \geq f(x) + \nabla f(x)^T(y-x)$	Local min = global min
Second-order cond.	$\nabla^2 f(x) \succeq 0$	Hessian PSD everywhere
Jensen's inequality	$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$ for convex f	Foundation of variational methods
Pointwise supremum	$\sup_\alpha f_\alpha(x)$, each f_α convex	Preserves convexity
Partial minimization	$\inf_y f(x,y)$, f jointly convex	Preserves convexity
Fenchel conjugate	$f^*(y) = \sup_x(y^T x - f(x))$	Always convex; $f^{**} = f$ if f closed convex
Fenchel's inequality	$f(x) + f^*(y) \geq x^T y$	Equality iff $y \in \partial f(x)$
Quasiconvex	$f(\theta x + (1-\theta)y) \leq \max\{f(x), f(y)\}$	Sublevel sets convex
Log-concave	$\log f$ concave, $f > 0$	Closed under products, marginalization
Log-convex	$\log f$ convex, $f > 0$	Implies convexity
Separating hyperplane	$\exists a \neq 0 : a^T x \leq b \leq a^T y$	Holds for disjoint convex sets
Supporting hyperplane	$a^T x \leq a^T x_0$ for all $x \in C$	Exists at every boundary point